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Borsa İstanbul Review

Borsa İstanbul Review 22-S2 (2022) S128–S140

http://www.elsevier.com/journals/borsa-istanbul-review/2214-8450

Full Length Article

Does ESG performance affect the financial performance of environmentally sensitive industries? A comparison between emerging and developed markets

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Received 17 August 2022; revised 19 November 2022; accepted 19 November 2022
Available online 1 December 2022

Abstract

This study seeks to investigate whether the environmental, social, and governance (ESG) performances of corporations from environmentally sensitive industries affect their financial performance. Moreover, the study determines how the impact of ESG performance on the corporate financial performance of environmentally sensitive corporations differs between developed and emerging countries. Our sample comprises 383 environmentally sensitive corporations, with 305 and 78 corporations from developed and emerging countries, respectively. Our findings indicate that overall ESG performance of environmentally sensitive corporations has a significantly positive relationship with the return on equity and Tobin's Q of the corporations. Moreover, our findings indicate that the impacts of the ESG performance of environmentally sensitive corporations on the financial performance are stronger in developed countries than emerging countries. This study contributes to the literature by adding a better understanding of the ESG–financial performance relationship, especially for the environmentally sensitive industries from both developed and emerging countries.
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JEL classification: G30; M14; M40

Keywords: ESG performance; Environmentally sensitive industry; Corporate social responsibility; Financial performance

1. Introduction

Corporate sustainability has become increasingly popular and influential in recent years for researchers and industries. With the growing global population and the rising climate change risk, the world is currently confronted with various environmental and social problems. To address these issues, strategies and initiatives related to corporate sustainability are critical for businesses today, particularly those whose success is more likely dependent on their sustainability initiatives and practices (Miralles-Quirós, Miralles-Quirós, & Gonçalves, 2018; Peylo, 2012). Several business organizations are concentrating on sustainability practices including waste management, carbon reduction, and improving their operational capability (Eccles & Serafeim, 2013). However, achieving corporate sustainability requires that these sustainability practices must benefit all stakeholders, while also increasing the corporation's value.

Stakeholder theory (Freeman, 1984) suggests that organizations should not only focus on generating value and profit for owners or stockholders but should also consider the best interests of all the stakeholders engaged with the business. Moreover, investing in ESG activities allows a business to achieve a competitive market advantage and results in improved financial performance (Freeman, 1984). Conversely, shareholder theory (Friedmann, 1962) opposes this stance, arguing that socially responsible activities are not a

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Peer review under responsibility of Borsa İstanbul Anonim Şirketi.

https://doi.org/10.1016/j.bir.2022.11.014
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corporation's responsibility or objective. Shareholder theory indicates that corporations should prioritize maximizing owner's value and profit and should conduct business operations only for the benefit of their stockholders or owners (Friedman, 1970) and that socially responsible activities are the responsibility of the relevant shareholder, not the corporation. However, shareholder theory does not restrict socially responsible practices as long as they are economically advantageous to the corporation (Smith, 2003), and previous studies found evidence showing ESG investments as being economically advantageous for the corporation (Huang, 2021).

The impact of ESG strategies and operations on corporate financial performance has been a topic of discussion in modern academic and business research. However, studies in ESG literature regarding the impact of ESG activities and performance of environmentally sensitive companies on their financial performance remain insufficient. Corporations operating in ecologically sensitive sectors are critical in terms of preserving ecological sustainability. Corporations in the energy, mining, metals, construction, chemical, and pulp and paper industries could be classified as environmentally sensitive owing to their production and business operations and environmental impact (Garcia, Mendes-Da-Silva, & Orsato, 2017). These corporations substantially invest on environmental management to mitigate the environmental issues created by their business operations. Moreover, these corporations may face stricter laws and regulations from the environment protection agencies and social pressure compared to nonsensitive corporations because of their nature of business operation (María Miralles-Quirós et al., 2018). This study investigates the ESG–financial performance relationship of environmentally sensitive corporations to determine how investment and expenditure made by these corporations to protect environment and communities affect their corporate financial performance. Moreover, this study seeks to examine whether the ESG operations and performance scores of environmentally sensitive corporations have a positive and value-adding relationship with the corporate financial performance by supporting the notions of stakeholder theory or these ESG operations and performance minimizes the shareholders' wealth by negatively affecting the financial performance of the corporation as claimed by the shareholder theory.

Furthermore, both developing and emerging countries now have institutional frameworks for ESG practice and reporting, but emerging countries lack adequate legislation, penalties, and incentives for ESG practices and efforts. Developed countries are more concerned with economic and environmental sustainability and sociocultural rights. Businesses operating in environmentally sensitive industries face increased pressure from stakeholders and government regulation and to incorporate ESG schemes into their routine business strategies. Corporations in emerging economies now voluntarily engage in more ESG practices, similar to corporations in developed markets, indicating that they receive greater economic reward from these voluntary ESG activities (Yoon, Lee, & Byun, 2018). The business environment, risks, and opportunities are considerably different between markets in developed and emerging economies. Environmentally sensitive corporations operating in developed and emerging economies market face different level of legislation and social pressure and avail different advantages in planning, executing, and reporting their ESG activities and operations. Owing to insufficient attention from ESG literature, it remains unclear if any significant difference can be found in the ESG–financial performance relationship between corporations in developed economies markets and corporations in the emerging economies market. Hence, this study seeks to investigate the financial value of ESG performances for environmentally sensitive corporations and the difference in the impact of ESG performance of these corporations between developed and emerging countries.

Additionally, ESG-related controversies are also receiving increasing attention from investors and policy makers. ESG controversies include adverse news or stories on questionable social conduct or environmentally harmful waste disposal, which draw media attention to corporations and, hence, cause concern from stakeholders and investors. Controversial news or information could impact corporation value as it would reflect suspicion on the corporation's long-term success (Aouadi & Marsat, 2018). Moreover, as part of an environmentally sensitive industry, corporate financial performance can be affected by controversial or adverse news regarding the corporation's environmental and business sustainability and social responsibility. This study also seeks to determine how ESG controversies affect the corporate financial performance of environmentally sensitive corporations. Further, this study analyzes whether the impact of ESG controversies on corporate financial performance of environmentally sensitive corporations differs between developed economies' markets and emerging economies' market.

The remainder of this study is structured as follows. Section 2 presents a comprehensive literature review followed by hypothesis development. Section 3 discusses the research design, empirical models, and study methodology, along with sample selection, data collection process, and a detailed description of variables. Section 4 presents the descriptive statistics and empirical findings with discussion. Finally, Section 5 contains the conclusions and some recommendations and suggestions.

2. Literature review and hypothesis development

Adopting ESG strategies and initiatives incurs significant organizational costs. Investors and corporations need to know if nonfinancial expenditures (e.g., ESG investments) result in corporate financial gains. Recently, numerous studies have examined the relationship between ESG activities and financial performance. Most studies in the ESG literature report a positive and significant relationship between ESG activities and operations with corporate financial performance. ESG performance had favorable effects on corporate financial performance measures (Alshehhi, Nobanee, & Khare, 2018; Bhaskaran, Ting, Sukumaran, & Sumod, 2020; De Lucia, Pazienza, & Bartlett, 2020; Friede, Busch, & Bassen, 2015).

Conversely, studies by Duque-Grisales (2019), Landi and Sciarelli (2019), and Marsat and Williams (2011) found a

negative relationship for ESG performance with the corporate financial performance of corporations. The abovementioned studies' negative findings represent and support the shareholder theory and the shareholder expense theory. However, few studies on the ESG–financial relationship have found an insignificant or nonexistent relationship of ESG performance with financial performance and market valuation (Atan, Alam, Said, & Zamri, 2019; Lopez-de-Silanes, McCahery, & Pudschedl, 2020).

A few studies have examined the ESG–financial performance relationship of some corporations from environmentally sensitive industries. Pätäri, Jantunen, Kyläheiko, and Sandström (2012) reported a positive correlation between sustainability and social responsibility performance and the financial and market performance of energy corporations. Garcia et al. (2017) reported that in emerging countries, firms belonging to sensitive industries perform better in corporate sustainability and socially responsible performance compared to corporations belonging to nonsensitive sectors. However, the authors noted that companies with the best ESG performance may be less profitable. Additionally, corporations from industries generating the most environmental pollution tend to contribute less to social policies and projects (Bhaskaran et al., 2020). Yoon et al. (2018) reported that ESG strategies and practices contribute less to the value generation of corporations from environmentally sensitive industries than corporations from nonsensitive industries. Conversely, Eriksson and Asgodom (2019) found no conclusive difference in the value-creating benefit of ESG performance between sensitive and conventional firms.

Unlike corporations in non-environmentally sensitive industries, those in environmentally sensitive industries are encouraged to voluntarily disclose significantly greater ESG-related information in their annual reports; For corporations in environmentally sensitive industries, ESG reporting is highly valued in the market (Lim, Wilmshurst, & Shimeld, 2010; Mohammadi, Mardani, Khan, & Streimikiene, 2018). However, corporations in these industries may struggle to establish a reputation as an ESG-conscious organization and enhance relationships with stakeholders due to their environmentally unfriendly business production and supply chain operations. Moreover, corporations with manufacturing processes that have significant adverse effect on the environment and corporations not considered to be environmentally sustainable are frequently excluded from negative screening when evaluating their performance and contribution to ESG (Dufwa & Hammarström, 2015). As environmentally sensitive businesses publish a greater volume and depth of ESG data (Garcia et al., 2017; Pled & Iatridis, 2012), these companies' ESG performances may have a greater impact on their market valuation and overall financial performance. Furthermore, as environmentally sensitive corporations voluntarily invest and engage in ESG operations, we expect that ESG expenditures will have either a positive impact on their financial performance as per stakeholder theory or a negative financial impact supporting the shareholder theory. Do expenditures made on ESG activities and operations by environmentally sensitive corporations produce any financial advantages? To explore organizations' financial performance, we will examine the relationship between ESG performance and profitability, operational efficiency, and market valuation. This study seeks to statistically determine whether the ESG operations and expenditure of environmentally sensitive corporations positively or negatively affect corporate financial performance. Hence, we present the following directional hypothesis:

Hypothesis 1A. Overall ESG performance of corporations from environmentally sensitive industries affects their profitability and market valuation

Over the last two decades, ESG practices and implications have attained widespread popularity in developed countries. Owing to the increased focus on sociocultural rights, economic and environmental sustainability, and other stakeholder concerns, corporations operating in developed economies need to manage the increased pressure and regulation from governments and other stakeholders to incorporate ESG schemes in their business strategies. Conversely, corporations in emerging countries can spend relatively less on ESG operations. ESG practice remains in its early stages in these countries, whereas corporations in developed countries need to heavily invest in ESG strategies (Manrique & Martí-Ballester, 2017). Due to differences in business environment and legislation, the implication and impact of corporate ESG performance on financial performance also varies between developed and emerging countries. Retaining environmental and social sustainability is equally critical for both developed and emerging countries. Regulatory authorities of both regions are encouraging corporations to implement sustainability-related practices. Environmentally sensitive corporations from both developed and emerging countries engage in and report ESG operations and activities. However, do developed and emerging countries differ on how corporate ESG performance affects their financial performance? Moreover, this study seeks to determine whether ESG operations and activities and ESG–financial performance relationships can increase or decrease the corporate financial value of environmentally sensitive corporations operating in developed or in emerging economies. Moreover, a great number of studies have been conducted on the value-enhancing capability of ESG performance in the developed countries' market in the ESG literature; however, studies focusing on emerging countries' markets are limited. The motivation behind investing in ESG operations is almost identical for corporations operating in both developed and emerging economies markets. We anticipate that the ESG performance of environmentally sensitive corporations affect their financial performance in both developed and emerging countries. In this study, we seek to determine whether the ESG performance of environmentally sensitive corporations differs in developing and emerging countries. Hence, we hypothesize the following:

Hypothesis 2A. ESG performance of corporations from environmentally sensitive industries in the developed

countries' markets affects their profitability and market valuation.

Hypothesis 2B. ESG performance of corporations from environmentally sensitive industries in the emerging countries' markets affects their profitability and market valuation.

Furthermore, previously published research (Aouadi & Marsat, 2018; Ting, Azizan, Bhaskaran, & Sukumaran, 2019) examined the effects of the ESG-related controversies of corporations and found a positive correlation between the ESG controversies score and corporations' valuation and financial performance. ESG controversy scores assess a corporation's exposure to environmental-, social-, and governance-related negative incidents in worldwide media. These controversies include harmful waste disposal, human rights violations, unethical practices of managers or CEOs, legal disputes, or company lawsuits. Negative news or scandals related to corporate sustainability or social responsibility issues may affect financial performance by affecting their goodwill as sustainable and socially responsible among the stakeholders. In this study, we examine whether these ESG controversy scores have a significant impact on financial performance and whether market participants react to unfavorable publicity or scandals involving ESG initiatives and the performance of environmentally sensitive corporations. Moreover, we seek to compare how ESG controversies on corporate financial performance differ by developed and emerging countries' market. Therefore, we hypothesized the following:

Hypothesis 3A. ESG controversies of corporations from environmentally sensitive industries affect their profitability and market valuation.

Owing to growing awareness on corporate social sustainability and accountability, negative reputation or controversial incidents related to the ESG of a corporation can be instrumental in the investment decisions of both individual and institutional investors. This study seeks to analyze whether controversial or negative news related to ESG operations of environmentally sensitive corporations affect their market valuation and financial performance.

3. Data description and methodology

3.1. Data source

This study examines the relationship between the ESG and financial performances of environmentally conscious organizations using the ESG performance scores provided by Refinitiv Eikon. Refinitiv Eikon's ESG score evaluates a corporation's ESG operations and performance using publicly available data (e.g., annual reports, corporate websites, stock market filings, news, nongovernmental organization websites, and corporate social responsibility [CSR] reports of respected corporations). Refinitiv Eikon's (formally known as Thomson Reuters Eikon) ESG scoring utilizes 186 essential measurements and contents related to sectors and industries to assess and score corporations' social and environmental performance.

Along with ESG data and information, Refinitiv Eikon provides a broad range of financial data and statistics for corporations worldwide. For this study, we collected financial data from Refinitiv Eikon and Refinitiv DataStream. Owing to the complexities involved, the relationship between ESG and financial performance will be evaluated using market-based performance and accounting-based measures.

3.2. Data sample

Corporations consuming high amounts of natural resources in their manufacturing processes and have negative impact on ecological sustainability have been considered samples for this study. This sample includes ESG data from corporations in six environmentally sensitive industries based on their manufacturing and operational activities, which are also known for their influence on environmental and socioeconomic sustainability (Garcia et al., 2017; Halme & Huse, 1997). First, we extracted ESG and financial performance data of 550 environmentally sensitive corporations from the Refinitiv database. However, owing to inconsistent and missing data, we needed to exclude 167 corporations from our sample. Our final sample includes 103 corporations from the energy sector, 85 corporations from heavy machineries and equipment, 76 from utilities, 40 from chemicals, 64 from mining and construction, and 15 corporations from pulp, paper, and packing sectors.

This study primarily seeks to assess the effect of ESG performance on the financial performance of environmentally sensitive industries that operate in developed and emerging countries based on country classifications established by the United Nations. The G-7 countries are considered representative of developed economies as they are recognized as "major developed economies" by the United Nations' World Economic Situation and Prospects report (WESP-UN, 2019). Hence, we included G-7 countries (i.e., Canada, France, Germany, Italy, Japan, the UK, and the USA) as developed countries in our sample. We included 16 major emerging economies from different world regions in our sample as emerging or developing countries, based on the UN's World Economic Situation and Prospects report (WESP-UN, 2019) and on their development growth rate, gross domestic product, and gross national income. The group for the sample population included Brazil, Russia, India, China, South Africa, South Korea, Mexico, Malaysia, Chile, Turkey, Indonesia, Argentina, Qatar, Nigeria, Thailand, and the United Arab Emirates for a 10-year sample. We used corporate ESG performance data from 2009 to 2018 to analyze and observe the relationship of corporate sustainable and socially responsible operations with the financial performance of those corporations. As per the previous research, we used a 1-year lag for the financial performance data to assess the appropriate impact of ESG performance on corporate financial performance and to eliminate the endogeneity issue (Atan et al., 2019; Manrique & Martí-Ballester, 2017; Velte, 2017). By

including a 1-year lag for financial performance data, we can better analyze the impact of ESG operations and performance of environmentally sensitive corporations on corporate financial performance. Therefore, we retrieved financial data for each environmentally sensitive corporation from 2010 to 2019 from the Refinitiv database. We excluded Argentina, Qatar, Nigeria, Thailand, and the United Arab Emirates from our emerging economies sample owing to a lack of data on the ESG performance of environmentally sensitive corporations in these countries. Our primary dataset contained data of 406 environmentally sensitive corporations from developed economies and from 144 environmentally sensitive corporations located and operating in leading emerging countries market worldwide. However, due to missing and inconsistent ESG data in the Refinitiv database, we eliminated 101 and 66 corporations from the developed and emerging economies, respectively, from our overall sample. After eliminating and removing corporations with missing ESG or financial performance data, our final sample contains 3830 observations from 305 to 78 corporations in the developing and emerging countries markets, respectively. Additionally, we sorted and organized the sample observations as a balanced panel data set to conduct a thorough analysis of the relationships between the corporation's performances over time. This is because panel data analysis is more rational and acceptable than individual data analysis in this case.

3.3. Independent variables

To analyze the relationship between ESG performance scores and financial performance measurements of the corporations, we conduct regression analyses. We used ESG performance scores as independent variables in the regression to observe how it affects our model's dependent variables (i.e., the financial performance measurements of the corporations in our sample). We discussed the variables of the study briefly below.

ESG Score: Refinitiv Eikon's overall ESG score (100%) is calculated using a weighted average of 34%, 42%, and 24% for environmental, social, and governance performance based on the latest ESG score calculation methodology (Environmental, Social and Governance (ESG) Scores from Refinitiv, 2022, p.20). In this study, we used Refinitiv Eikon's ESG score (aggregate) to indicate the overall ESG performance of corporations from environmentally sensitive industries.

Environmental score: Environmental score reflects a corporation's commitment toward environmental and ecological stability and sustainability. Refinitiv Eikon divides environmental scores into three major themes: resource use, emissions, and innovation. Refinitiv Eikon uses 68 different metrics to collect and measure the environmental scores of the corporations (Environmental, Social and Governance (ESG) Scores from Refinitiv, 2022, p.6).

Social score: Social score measures and describes how successful the corporation is in maintaining good relations and obtaining loyalty and trust from its employees, suppliers, consumers, and overall local community. Refinitiv Eikon divides the social score into four different main groups: workforce, human rights, community, and product responsibility. Refinitiv Eikon utilizes 62 different measures to gather and calculate the social score of corporations from their publicly available reports (Environmental, Social and Governance (ESG) Scores from Refinitiv, 2022, p.6).

Governance score: ESG governance seeks to ensure that corporations act in accordance with the best interests of its owners and ensure reporting transparency. For data collection and the scoring process, Refinitiv Eikon divided governance performance into three sub-themes: management, shareholders, and CSR strategy score. Refinitiv Eikon's calculates governance score by gathering and combining scores of 56 metrics from the public disclosures of corporations (Environmental, Social and Governance (ESG) Scores from Refinitiv, 2022, p.6).

ESG Controversies score: Refinitiv Eikon provides the ESG controversies score, which indicates whether the corporation experienced any controversies or scandals linked to ESG activities or operations in a given year. Refinitiv Eikon calculates the scores of each corporation's ESG controversies using 23 different ESG scandal or issue topics (Environmental, Social and Governance (ESG) Scores from Refinitiv, 2022, p.7). Refinitiv Eikon compiles information on a corporation's current controversies and scandals from all online and offline media sources, nongovernmental organization websites, stock markets, and other news sources. Moreover, Refinitiv Eikon also uses severity weighting to address market-cap bias. Larger firms acquire greater positive/negative media attention compared to smaller corporations (Refinitiv, 2021). Controversy scores are 0 and 100 by default and minimum, respectively; corporations with no controversies receive a score of 100.

3.4. Dependent variables

In this study, we used return on assets (ROA), return on equity (ROE), and Tobin's Q as dependent variables and proxies for the financial performance of corporations in line with previous studies (Bhaskaran et al., 2020; De Lucia et al., 2020; Shakil, Mahmood, Tasnia, & Munim, 2019; Ting et al., 2019; Velte, 2017).

Return on assets (ROA): ROA is an accounting-based measurement that can be used to describe a corporation's financial performance. Moreover, it indicates how successfully and efficiently a business utilizes its total assets in production and operation processes to produce profit and, hence, reflects business' operational performance.

Return on equity (ROE): ROE is an accounting-based measure that reflects a corporation's financial performance. Moreover, ROE is a profitability ratio that represents the financial performance and revenue-producing capability of the corporation.

Tobin's Q: Tobin's Q is a market-based measure for evaluating a corporation's financial performance and market value. Tobin's Q evaluates a corporation's financial performance by comparing its present market value to its book value or replacement value of total assets. Alongside market performance, Tobin's Q considers the long-term replacement cost of the corporation's total assets, which is important in sustainable investment practices.

3.5. Control variables

We included control variables as corporation characteristics in our models in line with previous studies to avoid biased assessments and obtain the actual impact of ESG performance on the corporate financial performance of environmentally sensitive corporations. In this study, control variables included the size and leverage of corporations. Large corporations encounter pressures from stakeholders for adopting sustainable and socially responsible practices (Rettab, Brik, & Mellahi, 2009). Additionally, big corporations obtain benefits of economies of scale that play a key role in financial performance. We have included the natural logarithm of total corporate assets to proxy for corporation size in the line with prior similar studies (Bhaskaran et al., 2020; Galbreath, 2013; Garcia et al., 2017; Laskar, Chakraborty, & Maji, 2017; Manrique & Martí-Ballester, 2017; Ting et al., 2019; Velte, 2017). For leverage, or the corporation's debt ratio, regular debt payments may help keep management effective. However, excessive leverage, high interest rates, or debt payments may result in lower cash flow for additional investments (e.g., ESG initiatives). To assess more relevant outcomes, we included the debt-to-equity and debt-to-assets ratios of environmentally sensitive corporations as leverage in our regression model, consistent with previous research (Ting et al., 2019; Velte, 2017).

3.6. Estimation procedure

To evaluate possible relations between the ESG performance of environmentally sensitive corporations and their financial performance and respond to the study's hypotheses, we developed and used two regression models. As ESG performance does not instantly affect a corporation's financial performance (Choi & Wang, 2009), regression models in this study included a 1-year lag for corporate financial performance measuring proxies. The use of a 1-year lag for financial performance measurements also enables more effective assessment of the effect of a corporation's sustainable and socially responsible performance on its financial performance (Atan et al., 2019; Manrique & Martí-Ballester, 2017; Velte, 2017). Hence, we compared the financial performance of corporation i at time t to their ESG performance at time $t-1$ to evaluate the effect of corporate sustainability and corporate social responsibility on the financial performance of environmentally sensitive corporations. We present the regression models as follows:

Model 1: This regression model seeks to examine the impact of overall ESG performance score on corporate financial performance:

$$ROA_{it} = \alpha + \beta_1 ESGScore_{it-1} + \beta_2 ControlVariables_{it} + \varepsilon_{it}; \tag{1.1}$$

$$ROE_{it} = \alpha + \beta_1 ESGScore_{it-1} + \beta_2 ControlVariables_{it} + \varepsilon_{it}; \text{ and} \tag{1.2}$$

$$Tobin'sQ_{it} = \alpha + \beta_1 ESGScore_{it-1} + \beta_2 ControlVariables_{it} + \varepsilon_{it}. \tag{1.3}$$

Where ROA_{it} comprises the returns on assets of corporation i at the time t;

ROE_{it} refers to the ROE of corporation i at time t;

$Tobin'sQ_{it}$ presents the market-based measurement of financial performance of corporation i at time t;

$ESGScore_{it-1}$ in all the models above represents overall ESG performance score of corporation i at the time $t-1$;

$ControlVariables_{it}$ refers to control variables, which are natural logs of total assets as the size of the corporation, and debt to equity and to assets as the leverage of corporation i at the time t; and

ε_{it} indicates the error term in the regression model.

Model 2: This regression model investigates the association of the performance of each individual pillar of ESG and the link of ESG controversies with the corporate financial performance of environmentally sensitive corporations.

$$\begin{aligned} ROA_{it} = {} & \alpha + \beta_1 Environmental_{it-1} + \beta_1 Social_{it-1} \\ & + \beta_1 Governance_{it-1} + \beta_1 ESG\ Controversies_{it-1} \\ & + \beta_2 ControlVariables_{it} + \varepsilon_{it}; \end{aligned} \tag{2.1}$$

$$\begin{aligned} ROE_{it} = {} & \alpha + \beta_1 Environmental_{it-1} + \beta_1 Social_{it-1} \\ & + \beta_1 Governance_{it-1} + \beta_1 ESG\ Controversies_{it-1} \\ & + \beta_2 ControlVariables_{it} + \varepsilon_{it}; \text{and} \end{aligned} \tag{2.2}$$

$$\begin{aligned} Tobin'sQ_{it} = {} & \alpha + \beta_1 Environmental_{it-1} + \beta_1 Social_{it-1} \\ & + \beta_1 Governance_{it-1} + \beta_1 ESG\ Controversies_{it-1} \\ & + \beta_2 ControlVariables_{it} + \varepsilon_{it} \end{aligned} \tag{2.3}$$

In all the above models, $Environmental_{it-1}$ represents the environmental performance score of corporations i at the time $t-1$;

$Social_{it-1}$ represents the social performance score of corporations i at the time $t-1$, $Governance_{it-1}$ represents the corporate governance performance score of corporation i at the time $t-1$;

$ESG\ Controversies_{it-1}$ represents the ESG scandal or controversies score of corporation i at the time $t-1$;

$ControlVariables_{it}$ refers to control variables, which are natural log of total assets as the size of the corporation, and debt to equity and to assets as the leverage of corporation i at the time t; and

ε_{it} indicates the error term in the regression model.

To begin, regression analysis will be conducted on the entire sample to evaluate the relationship between ESG and financial performance of environmentally sensitive corporations. Additionally, to compare the effect of ESG performance on the financial performance of corporations from developed and emerging countries, the regression models will be run separately for the samples from developed and emerging countries.

Table 1 presents the variables of our regression models and their descriptions.

Table 1
Regression variables.

Variables		Description	Period
Independent variables	ESG Score	Aggregate ESG score of the corporation as calculated by Refinitiv Eikon ESG database	2009 to 2018
	Environmental	Environmental performance score of the corporation by Refinitiv Eikon.	2009 to 2018
	Social	Social performance score of the corporation by Refinitiv Eikon.	2009 to 2018
	Governance	Corporate governance performance score of the corporation by Refinitiv Eikon.	2009 to 2018
	ESG Controversies	ESG related scandals and controversies of the corporation scored by Refinitiv Eikon.	2009 to 2018
Dependent variables	ROA	$\frac{\textit{Net Income before financing costs}}{\textit{Total Assets}}$	2010 to 2019
	ROE	$\frac{\textit{Net Income}}{\textit{Total Equity of Common Shares}}$	2010 to 2019
	Tobin's Q	$\frac{\textit{Market value of the corporation}}{\textit{Total Assets}}$	2010 to 2019
Control variables	Size	Natural log of the total assets of the corporations.	2010 to 2019
	Leverage	$\frac{\textit{Total Debt}}{\textit{Total Equity}}$	2010 to 2019
		$\frac{\textit{Total Debt}}{\textit{Total Assets}}$	2010 to 2019

Statistical analyses in this study have been conducted using the STATA 13 software. We present the empirical findings in the following section.

4. Empirical analyses and findings

4.1. Descriptive statistics

The final panel data set for this study contains the ESG and financial data of a total of 383 unique corporations from 7 developed and 11 emerging countries for a 10-year period. Table 2 summarizes the descriptive statistics of our dependent, independent, and control variables as follows:

Fig. 1 below presents the ESG performance scores of the corporations by industry and inter-industry differences in ESG performance scores of the corporations.

As the graph indicates, the average ESG performance scores of corporations from environmentally sensitive industries in our sample are nearly similar. Moreover, it provides the impression that most firms are satisfied with implementing just enough ESG practices and processes to maintain market competitiveness.

Furthermore, we have conducted F-, Breusch–Pagan LM, and, consequently, Hausman tests to select the appropriate models for the regression analysis of the study. We conducted regressions with Driscoll and Kraay standard errors to eliminate heteroscedasticity, cross-sectional dependency, and first-order autocorrelation in our sample. Driscoll and Kraay (1998) standard errors are heteroscedastic, and their estimation is expected to be resilient to autocorrelation and cross-sectional dependency (Hoechle, 2007).

Table 3 below represents correlation among the variables in our sample.

4.2. Regression results

As the regression results suggest, the ESG score, which represents overall ESG performance, has no significant relationship with the ROA of environmentally sensitive corporations from developed and emerging countries (Table 4). The

Table 2
Summary Statistics (overall sample).

Variable	N	Mean	Std. Dev.	Median	Min.	Max.	Skew.	Kurt.
ESG Score	3830	49.072	20.196	49.34	1.24	92.11	−0.11	2.249
Environmental	3830	48.825	26.392	50.845	0.00	97.15	−0.215	2.009
Social	3830	45.561	23.634	43.245	0.05	98.11	0.151	2.084
Governance	3830	54.237	22.677	56.34	1.24	98.73	−0.222	2.049
ESG Controversies	3830	88.486	24.728	100	0.88	100	−2.158	6.413
ROA	3830	0.033	0.847	0.04	−31.25	32.33	−4.156	1168.91
ROE	3830	0.094	0.595	0.11	−29.75	1.39	−41.64	1927.20
Tobin's Q	3830	0.869	0.892	0.61	0.01	9.58	3.507	21.01
LN_Total Assets	3830	23.062	1.419	22.98	15.9	26.74	−0.122	3.652
Debt to Equity	3830	1.091	2.224	0.64	0.00	50.02	11.141	176.85
Debt to Assets	3830	0.337	1.261	0.26	0.00	24.95	19.027	368.94

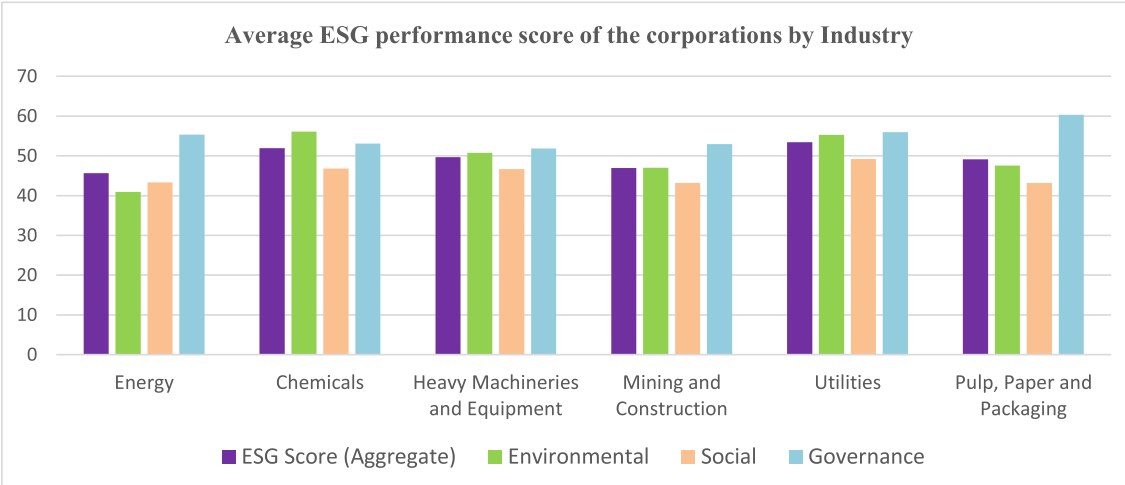

Fig. 1. ESG performance (average) scores of the corporations by industry.

ESG operations of these corporations may not be enough to affect ROA because accounting financial performance measurements like ROA usually combine a larger range of corporate activity than ESG operations in the corporations (Huang, 2021). However, overall ESG performance score contains a positive and significant relationship with corporate ROE, which suggests that ESG performance can positively affect the profitability and the financial performance of environmentally sensitive corporations. Similarly, Tobin's Q of environmentally sensitive corporations has a positive and significant relationship with the overall ESG performance score. This indicates that ESG performance could positively contribute to the market valuation of the corporations. From theoretical perspective, the positive significant relationship of ROE and Tobin's Q with the overall ESG performance of environmentally sensitive corporations supports the stakeholder theory (Freeman, 1984) and opposes the shareholder theory (Friedmann, 1962).

Table 5 indicates that the environmental scores of the corporations have a positively significant relationship with the ROA only, which implies that increase of environmentally sustainable operations such as improved emission, resource use, and innovation scores positively affect their profitability. Furthermore, ROE is significantly and positively correlated with the governance performance of the corporations. This suggests that better and effective governance practice leads to higher ROE for the corporations. The social performance score of environmentally sensitive corporations alone has significant

Table 3
Pearson correlation matrix.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
(1) ESG Score	1.000										
(2) Environmental	0.881*	1.000									
	(0.000)										
(3) Social	0.893*	0.693*	1.000								
	(0.000)	(0.000)									
(4) Governance	0.632*	0.340*	0.416*	1.000							
	(0.000)	(0.000)	(0.000)								
(5) ESG Controversies	−0.281*	−0.227*	−0.293*	−0.147*	1.000						
	(0.000)	(0.000)	(0.000)	(0.000)							
(6) ROA	0.034*	0.035*	0.031	0.008	−0.007	1.000					
	(0.048)	(0.038)	(0.066)	(0.639)	(0.682)						
(7) ROE	0.061*	0.053*	0.068*	0.015	−0.017	0.465*	1.000				
	(0.000)	(0.002)	(0.000)	(0.381)	(0.314)	(0.000)					
(8) TobinsQ	−0.112*	−0.183*	−0.036*	−0.035*	0.094*	0.027	0.070*	1.000			
	(0.000)	(0.000)	(0.036)	(0.041)	(0.000)	(0.097)	(0.000)				
(9) LN_TotalAssets	0.486*	0.482*	0.408*	0.259*	−0.326*	0.032	0.054*	−0.327*	1.000		
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.059)	(0.001)	(0.000)			
(10) DebtToEquity	0.046*	0.054*	0.031	0.019	−0.022	−0.001	0.013	−0.144*	0.153*	1.000	
	(0.004)	(0.001)	(0.052)	(0.229)	(0.180)	(0.972)	(0.451)	(0.000)	(0.000)		
(11) DebtToAssets	0.044*	0.030	0.042*	0.041*	0.006	0.000	0.001	−0.059*	0.092*	0.598*	1.000
	(0.006)	(0.063)	(0.010)	(0.010)	(0.703)	(0.977)	(0.939)	(0.001)	(0.000)	(0.000)	

***p < 0.01, **p < 0.05, *p < 0.1.

Table 4
Regression result (Model 1-overall sample).

Model and Dependent variable	Model 1.1 (ROA)			Model 1.2 (ROE)			Model 1.3 (Tobin's Q)		
Variable	Coefficient	Drisc/Kraay Stand. Error	p-value	Coefficient	Drisc/Kraay Stand. Error	p-value	Coefficient	Drisc/Kraay Stand. Error	p-value
ESG Score	0.00079	0.00050	0.147	0.00213	0.00046	0.001***	0.00141	0.00054	0.029**
LN_Total Assets	0.00960	0.00391	0.037**	−0.08697	0.01791	0.001***	−0.33185	0.02249	0.000***
Debt to Equity	−0.00133	0.00115	0.278	0.01156	0.00537	0.060*	0.01543	0.00564	0.023**
Debt to Assets	−0.00055	0.00157	0.733	−0.44214	0.17097	0.029**	−0.84713	0.27892	0.014**
Constant	−0.22586	0.00157	0.057*	2.13152	0.42862	0.001***	8.72183	0.52011	0.000***
No. of Obs.	3830			3830			3830		
R-squared	0.0010			0.0058			0.0664		

***p < 0.01, **p < 0.05, *p < 0.1.

Table 5
Regression result (Model 2-overall sample).

Model and Dependent variable	Model 2.1 (ROA)			Model 2.2 (ROE)			Model 2.3 (Tobin's Q)		
Variable	Coefficient	Drisc/Kraay Stand. Error	p-value	Coefficient	Drisc/Kraay Stand. Error	p-value	Coefficient	Drisc/Kraay Stand. Error	p-value
Environmental	0.00141	0.00049	0.017**	0.00059	0.00082	0.485	−0.00003	0.00043	0.948
Social	−0.00065	0.00042	0.155	0.00060	0.00048	0.239	0.00185	0.00054	0.008***
Governance	−0.00006	0.00101	0.956	0.00041	0.00014	0.017**	−0.00051	0.00029	0.105
ESG Controversies	0.00009	0.00013	0.503	0.000007	0.00016	0.965	0.00093	0.00021	0.001***
LN_Total Assets	0.00758	0.00920	0.431	−0.00003	0.00589	0.997	−0.33174	0.02202	0.000***
Debt to Equity	−0.00172	0.00212	0.436	0.00862	0.00685	0.240	0.01575	0.00577	0.023**
Debt to Assets	0.00026	0.00365	0.944	−0.01161	0.00685	0.124	−0.84906	0.27947	0.014**
Constant	−0.18518	0.26481	0.502	0.00983	0.15669	0.951	8.65130	0.51014	0.000***
No. of Obs.	3830			3830			3830		
R-squared	0.0014			0.0041			0.0689		

***p < 0.01, **p < 0.05, *p < 0.1.

and positive relationship with Tobin's Q of corporations in our overall sample. This suggests that the social strategies and actions of environmentally sensitive corporations are recognized and positively valued by the market and the higher level of social performance can contribute to the increase in the value of the corporations. To reiterate, no negatively significant relationship was found between ESG individual pillars' performance scores of the corporations and their financial performance indicators, which opposes the shareholder theory, and positive relationships support claims of stakeholder theory.

The positive and statistically significant relationship of ESG controversies score and Tobin's Q indicates that a higher ESG controversies score positively affects market valuation of environmentally sensitive corporations. The results indicate that market investors acknowledge scandals and controversies related to the ESG activities of environmentally sensitive corporations, and corporations with less-negative ESG-related news or events tend to have a better market valuation.

To analyze and compare how ESG performances of environmentally sensitive corporations from developed countries and emerging countries affect financial performance, we have investigated the ESG–financial performance relationship separately for corporations from developed and emerging countries. Table 6 presents the results of the regressions conducted with data of the corporations from developed countries. As the outcomes of the regression models suggest, the overall

Table 6
Regression result (Model 1-sample from developed countries).

Model and Dependent variable	Model 1.1 (ROA)			Model 1.2 (ROE)			Model 1.3 (Tobin's Q)		
Variable	Coefficient	Drisc/Kraay Stand. Error	p-value	Coefficient	Drisc/Kraay Stand. Error	p-value	Coefficient	Drisc/Kraay Stand. Error	p-value
ESG Score	0.00100	0.00063	0.145	0.00271	0.00056	0.001***	0.00122	0.00038	0.010**
LN_Total Assets	0.01046	0.00465	0.051*	−0.08612	0.02144	0.003***	−0.28270	0.03301	0.000***
Debt to Equity	0.00009	0.00080	0.910	0.01481	0.00742	0.077*	0.01254	0.00406	0.013**
Debt to Assets	−0.00157	0.00132	0.267	−0.58735	0.23138	0.032**	−0.74412	0.26168	0.019**
Constant	−0.26096	0.11792	0.054*	2.12566	0.51626	0.003***	7.56484	0.70363	0.000***
No. of Obs.	3050			3050			3050		
R-squared	0.0011			0.0062			0.0531		

***p < 0.01, **p < 0.05, *p < 0.1.

Table 7
Regression result (Model 2-sample from developed countries).

Model and Dependent variable	Model 2.1 (ROA)			Model 2.2 (ROE)			Model 2.3 (Tobin's Q)		
Variable	Coefficient	Drisc/Kraay Stand. Error	p-value	Coefficient	Drisc/Kraay Stand. Error	p-value	Coefficient	Drisc/Kraay Stand. Error	p-value
Environmental	0.00171	0.00056	0.014**	0.00081	0.00092	0.401	−0.00085	0.00057	0.171
Social	−0.00087	0.00044	0.077*	0.00068	0.00052	0.222	0.00280	0.00072	0.004***
Governance	0.00008	0.00126	0.949	0.00063	0.00012	0.001***	−0.00068	0.00025	0.025**
ESG Controversies	0.00006	0.00015	0.706	−0.00010	0.00017	0.552	0.00062	0.00036	0.123
LN_Total Assets	0.00768	0.01072	0.492	−0.00293	0.00639	0.658	−0.28316	0.06577	0.002***
Debt to Equity	−0.00052	0.00195	0.794	0.01231	0.00792	0.154	0.01288	0.00470	0.023**
Debt to Assets	−0.00041	0.00354	0.909	−0.01454	0.00779	0.095*	−0.73839	0.30180	0.037**
Constant	−0.20196	0.30961	0.531	0.05605	0.17499	0.756	7.53056	1.46662	0.001***
No. of Obs.	3050			3050			3050		
R-squared	0.0016			0.0055			0.0566		

***p < 0.01, **p < 0.05, *p < 0.1.

ESG performance score is positively and significantly associated with ROE and Tobin's Q for the corporations from developed countries' market.

We have examined the relationship of the ESG individual pillar's performance with the financial performance of environmentally sensitive corporations separately to analyze if performance score for any individual pillar has a significantly stronger correlation with the financial performance as compared to the overall ESG score. Table 7 presents the regression results on the impact of each pillar's performance and the effect of ESG controversies on the corporate financial performance of environmentally sensitive corporations from developed countries' markets. The results indicate that the governance performance score of corporations from developed countries has a significant relationship with ROE, suggesting the importance of good governance for corporations. The positive and significant relationship of social performance score and Tobin's Q expresses that corporation's social activities and contributions are valued and recognized in the markets of developed countries. Furthermore, the positive correlation between ROA and environmental performance score expresses that being environmentally sensible and responsible may ensure better ROA for corporations.

Interestingly, social performance scores of corporations from have a slight negatively significant relationship with their ROA. Governance score also has negative correlation with Tobin's Q, indicating the opposite of the notion of stakeholder theory. Moreover, although the governance performance score could positively affect ROE, it was not recognized and valued positively by the market. The results demonstrate the impact of individual pillar performance of ESG on profitability and market valuation measurement. The ESG controversies score of environmentally sensitive corporations from developed countries do not significantly affect any of the financial performance measures. Stakeholders in developed markets may be more sensitive about the overall ESG and individual pillars performance score than the ESG controversies score.

Table 8 presents the impact of overall ESG performance on the corporate financial performance of environmentally sensitive corporations from emerging countries' markets. As per the regression results, the overall ESG performance of environmentally sensitive corporations from the emerging countries do not have any significant impact on the financial performance measures ROA, ROE, and Tobin's Q.

According to the regression results (Table 9), the ESG score exhibits no significant relationship with the ROA, ROE, or

Table 8
Regression result (Model 1-sample from emerging countries).

Model and Dependent variable	Model 1.1 (ROA)			Model 1.2 (ROE)			Model 1.3 (Tobin's Q)		
Variable	Coefficient	Drisc/Kraay Stand. Error	p-value	Coefficient	Drisc/Kraay Stand. Error	p-value	Coefficient	Drisc/Kraay Stand. Error	p-value
ESG Score	−0.00010	0.00009	0.313	0.00014	0.00039	0.734	0.00283	0.00241	0.270
LN_Total Assets	−0.00985	0.00223	0.002***	−0.07760	0.00709	0.000***	−0.53092	0.09686	0.000***
Debt to Equity	−0.00310	0.00180	0.118	−0.00120	0.00429	0.785	0.03484	0.01396	0.034**
Debt to Assets	−0.08990	0.03748	0.040**	−0.11853	0.07980	0.172	−1.10448	0.29770	0.005***
Constant	0.31682	0.06834	0.001***	1.94902	0.17801	0.000***	13.39329	2.43384	0.000***
No. of Obs.	780			780			780		
R-squared	0.1157			0.0570			0.1172		

***p < 0.01, **p < 0.05, *p < 0.1.

Table 9
Regression result (Model 2-sample from emerging countries).

Model and Dependent variable	Model 2.1 (ROA)			Model 2.2 (ROE)			Model 2.3 (Tobin's Q)		
Variable	Coefficient	Drisc/Kraay Stand. Error	p-value	Coefficient	Drisc/Kraay Stand. Error	p-value	Coefficient	Drisc/Kraay Stand. Error	p-value
Environmental	−0.00023	0.00017	0.211	−0.00048	0.00032	0.172	0.00307	0.00251	0.253
Social	0.00017	0.00019	0.398	0.00037	0.00042	0.399	−0.00118	0.00348	0.741
Governance	−0.00003	0.00011	0.774	−0.00011	0.00017	0.517	0.00105	0.00214	0.634
ESG Controversies	0.00019	0.00010	0.097*	0.00024	0.00027	0.419	0.00215	0.00062	0.007***
LN_Total Assets	−0.00866	0.00388	0.052*	−0.01144	0.01269	0.391	−0.52610	0.11658	0.001***
Debt to Equity	−0.00285	0.00312	0.385	−0.00319	0.00608	0.611	0.03820	0.01445	0.027**
Debt to Assets	−0.09394	0.04278	0.056*	−0.11714	0.05829	0.075*	−1.15101	0.39377	0.017**
Constant	0.27249	0.10394	0.028*	0.03964	0.32332	0.251	13.09542	2.79012	0.001***
No. of Obs.	780			780			780		
R-squared	0.1355			0.0337			0.1251		

***p < 0.01, **p < 0.05, *p < 0.1.

Tobin's Q of environmentally sensitive corporations from emerging countries. However, ESG controversies score has a positive and slightly significant relationship with the ROA of these corporations and a positive and strong significant correlation with Tobin's Q. The positive and significant correlation indicates that a higher ESG controversies score leads to higher ROA and greater market valuation (Tobin's Q) for environmentally sensitive corporations from emerging countries. Growing interests of ESG among the general public and investors in the emerging markets could constitute a possible reason. Moreover, corporations in involved in ESG controversies could face regulatory actions from respective authorities, which may affect overall operation and valuation. That is why investors in these emerging countries may seem more sensitive to these ESG controversies rather than the individual or overall ESG performance score. Corporation's exposure to negative incidents related to ESG operations or controversial news regarding these issues negatively affect their market valuation. Moreover, these results indicate that by being environmentally and socially accountable and free from ESG related controversies, corporations can ensure efficient and profitable asset use.

To confirm the results of our models, an additional regression analysis was conducted as a robustness test. ESGC score, which is a combined ESG score of corporations with an overlay of ESG controversies scores of corporations, was considered an independent variable in this case and regressed against the same dependent variables of our models to determine how ESG performance affects corporations' financial performance. Additional regression analyses results confirm the main models of this study.

5. Conclusion

This study examines the relationship between the ESG performance and corporate financial performance of corporations from environmentally sensitive industries. We compare the ESG performance of environmentally sensitive industries between emerging and developed countries. Our findings indicate that the overall ESG performance of environmentally sensitive corporations have positive and significant relationships with ROE and Tobin's Q, indicating that higher ESG performance positively impacts profitability and contribute to the increase in market value of these corporations. From a theoretical perspective, this positive impact of ESG performance of environmentally sensitive corporations on their profitability and market valuation supports the stakeholder theory and opposes the shareholder theory. The positive impact of ESG performance on corporate financial performance of environmentally sensitive corporations suggests that investing in and expanding ESG operations and activities can bring financial benefits along with preserving environmental and social sustainability. However, the insignificant relationship between ROA and ESG performance does not support either stakeholder or shareholder theory, but legitimacy theory could be the reason why corporations engage in the ESG operations and activities. According to legitimacy theory, corporations plan and practice ESG activities to comply with ESG regulations of governments, to appease the external stakeholders, and to validate the presence of the corporation in the society. Corporations operating in developed and emerging markets have to face different laws and regulations and there are differences in the economical and sociopolitical conditions between these countries. Our findings suggest that ESG performances of environmentally sensitive corporations are more recognized in the developed countries' markets and have a stronger impact on the financial performance of the corporations from developed countries than on those from emerging countries. The ESG performance of environmentally developed corporations from developed countries has a positive relationship with corporate financial performance indicators whereas the ESG performance of the corporations from emerging countries does not have a significant relationship with their financial performance indicators. The possible explanation of the insignificant relationship of corporate financial performance with overall ESG performance could be that environmentally sensitive corporations operating in emerging economies' market may have gained their ESG performance scores by

engaging themselves in ESG activities to maintain legal obligations and social contracts, which did not have a significant impact on profitability or market valuation. Moreover, this may be because stakeholders and investors lack confidence and trust in corporations' ESG strategies and operations in the context of emerging countries, which may subsequently affect the profitability and market valuation of the corporation in a positive manner, as is the case in the developed countries. Conversely, transparent and comparatively more ethical business practice in developed countries could also be the cause of the stronger and positive effect of ESG performance of environmentally sensitive corporations on financial performance in developed countries as compared to emerging countries.

Furthermore, the findings show that the investors and the market recognize and react to the ESG-related controversies of environmentally sensitive corporations and the effect of ESG controversies score of environmentally sensitive corporations on financial performance is stronger and more noticeable in emerging countries than in developed countries.

The results of this study provide a better understanding of the impact of the ESG performance on the financial performance especially for the environmentally sensitive corporations from the legal and economic contexts of developed and emerging countries. The findings also corroborate the findings of existing academic research where most studies have found a positive link between ESG and financial performance (Alshehhi et al., 2018; Friede et al., 2015; Orlitzky, Schmidt, & Rynes, 2003). Moreover, this study contributes to the literature by focusing on the sustainability practices of business organizations and its financial impact. The separate analysis of the ESG–financial performance relationship of environmentally sensitive corporations from developed and emerging markets is an important addition that shows the difference in the impact of ESG practices in these countries. The findings of this study would help researchers who seek to investigate different region-based or multi country-based ESG–corporate financial performance relationships for corporations from different sectors. The results of this study make a significant contribution to the theoretical field of the financial value of ESG performance and to the ESG–financial performance relationship of corporations from environmentally sensitive industries.

The study finds that the high ESG performances of environmentally sensitive corporations positively affect profitability and may also increase market value by supporting stakeholder theory. However, the financial value of the ESG performances of environmentally sensitive corporations from emerging countries remain insignificant and inconclusive, suggesting that more emphasis is needed on ESG strategies and initiatives in these countries. The findings further contribute to the decision-making process regarding the environmental and social sustainability issues of environmentally sensitive corporations from developed and emerging countries. This comparison-based study would help international investors and regulatory bodies understand the impact of the ESG performance of environmentally sensitive corporations in different economic regions and make necessary investment decisions. Moreover, this study seeks to encourage the management of environmentally sensitive corporations to adopt more efficient and effective ESG policies and initiatives as ESG performance can maximize both profitability and market value.

Acknowledgement

Earlier version of the paper was presented at the 10th Istanbul Finance Conference held at Istanbul Ticaret University, Istanbul, Türkiye during 4–5 November 2021.

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